



UNIVERSITY OF THE PHILIPPINES

Master of Science in Materials Science and Engineering

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Investigation of a-Si/SiO₂ antireflective coatings for terahertz pump wavelengths

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CHAPTER 1

INTRODUCTION

The development of devices which operate in the terahertz range has been a recent area of interest for researchers. The applications of terahertz waves include analysis of biological specimens[1], pharmaceuticals[2], and imaging[3] among others. The mechanisms by which these devices operate may include upconversion from the radio and mmWave range, or downconversion of photonic devices from the infrared region. An example of a device which utilizes downconversion is the terahertz photoconductive antenna (PCA), and there are already multiple demonstrations and even commercial PCA devices that utilize the 780 nm and 1550 nm wavelengths, which are supplied by Ti:sapphire and telecommunication fiber lasers respectively. A PCA consists of a semiconductor material where the terahertz generation or absorption occurs and two metal electrodes on the semiconductor surface separated by a few microns that forms the photoconductive gap[4]. When a femtosecond pulse from the laser illuminates the gap, carriers are generated in the semiconductor and swept by an electric field from the bias applied to the metal electrodes, and the movement of the carriers generates the terahertz wave.

One of the common semiconductor materials of choice for a PCA is low temperature (LT) grown indium gallium arsenide (InGaAs), which has a bandgap of $E_g < 0.8$ eV and can be excited by common telecommunications wavelength lasers operating at 1550 nm [5]. Another common semiconductor material used for PCAs is LT gallium arsenide (GaAs), which has a bandgap of $E_g < 1.42$ eV and is operated at a much shorter wavelength of 780 nm. However, when a bare PCA is illuminated by a laser there are significant reflection losses due to the high refractive index of the material[6]. Adding an antireflective coating to the PCA can increase the efficiency of the device by reducing the amount of light reflected by the film. The challenge lies in the design of a

suitable antireflective coating for the aforementioned wavelengths.

1.1 Significance of the study

The development of terahertz (THz) PCAs which operate efficiently at commonly used wavelengths holds great potential for integrating THz technology with existing optical systems. By enhancing the efficiency of THz generation, this study contributes to the advancement of THz time domain spectroscopy (TDS) systems, which have diverse applications in fields such as biology, imaging, and pharmaceuticals. The Semiconductors group of the Condensed Matter Physics Laboratory in UP Diliman is able to grow LT-InGaAs and LT-InGaAs layers via molecular beam epitaxy (MBE), fabricate them into PCAs via photolithography, and study them using 1550 nm and 780 nm wavelength lasers. However, the PCAs being developed lack antireflective coatings optimized for their operating wavelengths. By integrating an antireflective coating (ARC) stack composed of a-Si and SiO₂, two materials commonly used for CMOS processing, this study aims to address these problems, improving the THz intensity generated by the PCA and its overall performance. The findings could pave the way for more efficient THz devices, enhancing capabilities in scientific research, medical diagnostics, and industrial processes.

1.2 Objectives of the study

The main objective of this study is to investigate the feasibility of using a-Si and SiO₂ thin films to create an antireflective coating which can be used for LT-InGaAs PCAs. Specifically, this will require the following objectives:

- to identify the optical properties of a-Si and SiO₂ thin films deposited using electron beam evaporation,
- to design an antireflective coating utilizing a-Si and SiO₂ optimized for the wavelengths at which LT-InGaAs and LT-GaAs THz PCAs can be operated, and

- to evaluate the reflectivity of the fabricated a-Si/SiO₂ antireflective coating at the relevant wavelengths.

1.3 Scope and limitations of the study

The study will only cover the investigation of an amorphous silicon (a-Si) and SiO₂ ARC for LT-InGaAs and LT-GaAsTHz PCA applications. As such, the study will not investigate the growth of the LT-InGaAs and LT-GaAs film using MBE and will also be limited to electron beam evaporation as the deposition method for the a-Si/SiO₂. The characterization is also limited to the evaluation of the reflectivity of the designed films and will not include implementations of the developed system.

CHAPTER 2

REVIEW OF RELATED LITERATURE

2.1 Terahertz photoconductive antenna

The range of frequencies between microwave and infrared in the electromagnetic spectrum is known as the THz range. It was also called the THz gap in the past due to the difficulty of finding emitters and detectors which worked in this frequency range[7]. THz radiation is non-ionizing and can penetrate through organic material such as plastic and clothing, thus it has potential use in biological characterization[1], detection of contraband materials[8], and chemical identification[9]. Some substances have specific THz absorption spectra which can be used to identify their composition, as shown in Figure 2.1. The interaction amongst the bonds and forces in the molecules result in vibrational modes visible in the THz region[8]. On the other hand, THz reflection can be used to detect concealed objects when the concealing material is transparent to THz radiation as in refthrough, as the case for dielectrics such as clothing[10]. A common setup for some of these applications involves THz TDS.

In THz TDS, a THz pulse is generated using a PCA. The shape of the electrodes influences the pattern of the THz radiation generated by the PCA. Dipole and bowtie

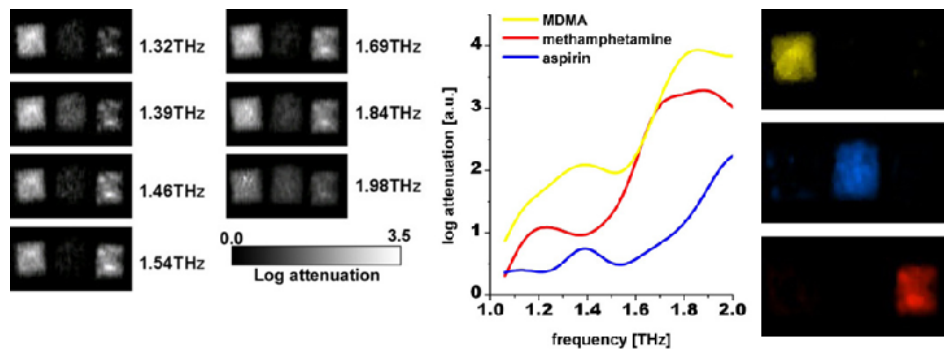


Figure 2.1: THz identification of MDMA, methamphetamine, and aspirin through a paper envelope. The substances are differentiated using their THz spectra[8]

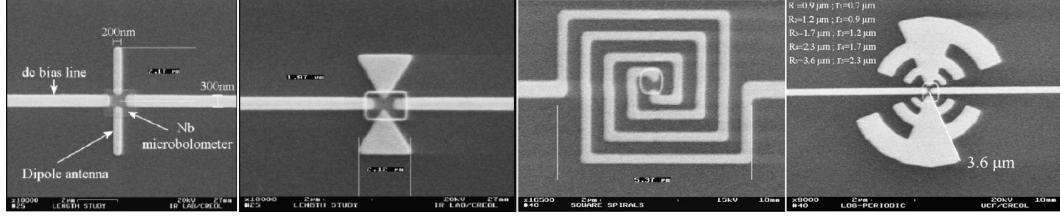


Figure 2.2: Commonly used THz PCA geometries; from left to right: dipole, bowtie, spiral, and log-spiral[11]

patterns demonstrate higher polarization ratios compared to spiral antennas while spiral and log-spiral antennas have a higher calculated radiation efficiency compared to bowtie and dipole antennas[11]. The shapes of the antennas are presented in Figure 2.2.

The emission mechanism for the PCA is as follows: the photoconductive material between the electrodes is illuminated with a femtosecond laser pulse, which generates photocarriers that are swept by an electric field applied to the electrodes of the PCA[4], [5]. The transient current $i_{PC}(t)$ resulting from the moving carriers generates electromagnetic radiation according to Maxwell's equations, with the intensity of the electric field component of the THz wave radiated by a dipole antenna expressed by Equation 2.1

$$E_{THz}(r, \theta, t) = \frac{l_e}{4\pi\epsilon c^2 r} \frac{\partial i_{PC}(t)}{\partial t} \sin(\theta) \quad (2.1)$$

where l_e is the antenna length, ϵ the permittivity, c the speed of light, r the distance from the dipole, and θ the angle relative to the dipole axis[4].

The transient current density term $i_{PC}(t)$ is proportional to the convolution integral of the carrier density $n(t)$ and electron velocity $v_e(t)$ as shown in Equation 2.2[4].

$$i_{PC}(t) \propto P_{pulse}(t) \otimes (n(t) v_e(t)). \quad (2.2)$$

The carrier density is described by

$$\frac{dn}{dt} = \eta \frac{P_{pulse}(t)}{h\nu} - \frac{n}{\tau_c} \quad (2.3)$$

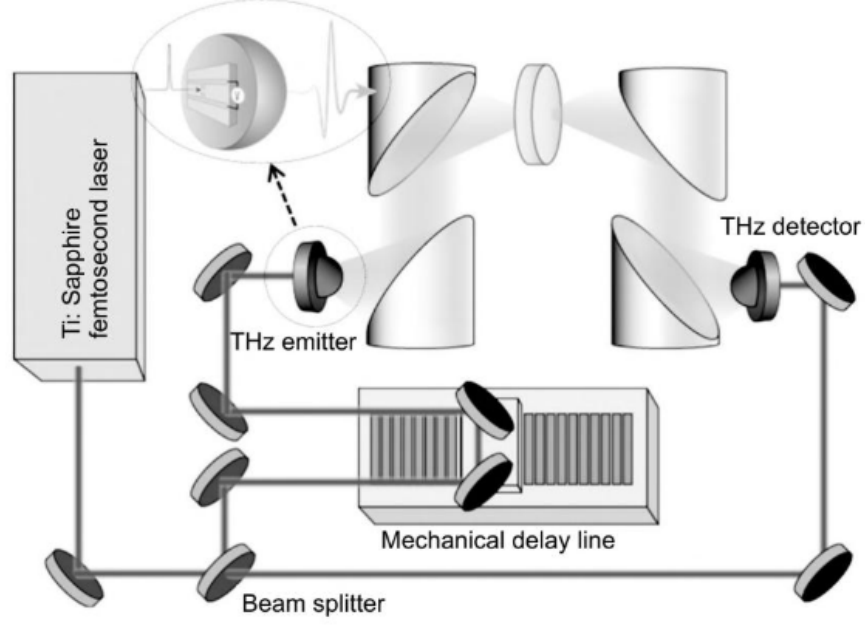


Figure 2.3: Schematic of a THz TDS setup with a femtosecond laser and two photoconductive antennas[4]

and using a 1D Drude-Lorentz model, the electron velocity by

$$\frac{dv_e}{dt} = -\frac{v_e}{\tau_s} + \frac{e}{m^*}E \quad (2.4)$$

where η is the absorption efficiency, $P_{\text{pulse}}(t)$ the ultrashort laser pulse power considered over time, τ_c , τ_s , and m^* the carrier lifetime, momentum relaxation time, electron effective mass respectively, and E the local electric field[4].

A photoconductive antenna is also used to detect THz radiation, with the incoming THz pulse focused together with a femtosecond pulse split from the same source at the gap of the antenna. As the femtosecond probe pulse arrives at the gap, carriers are excited and then accelerated proportionally by the electric field provided by the incoming THz pulse[5]. The difference of the probe pulse and the THz pulse provides a measurement in the time domain for the transient of the THz pulse. A schematic of a complete THz TDS system with emitter and detector is shown in Figure 2.3. Some of the photoconductive materials used for THz TDS systems include LT grown GaAs and InGaAs[13], [14], [15]. LT-GaAs is grown via molecular beam epitaxy at 200 °C to 300 °C instead of the usual growth temperature around 600 °C[16].

2.2 Electromagnetic Waves in Dielectric Materials

The propagation of electromagnetic waves through dielectric materials can be described using the Fresnel equations, which can be derived from the boundary conditions that constrain the electric and magnetic field components arising as a direct result of Maxwell's equations. Suppose incident light approaches a boundary between two dielectric materials with refractive indices n_1 and n_2 as shown in Figure 2.4. The first case shows s-polarized light, where the electric field is perpendicular to the plane of incidence, while the second case shows p-polarized light, where the electric field is parallel to the plane of incidence. The incident and reflected angles are equal $\theta_i = \theta_r$, and the transmitted angle is given by Snell's law $n_1 \sin \theta_i = n_2 \sin \theta_t$. The boundary conditions require that the components of the electric and magnetic fields be continuous, thus for s-polarized light, the electric fields are given by $E_i + E_r = E_t$ or $1 + r = t$, where r and t are the reflected and transmitted amplitudes and the incident amplitude is normalized to 1. The y-component of the magnetic field is given by $H_i \cos \theta_i - H_r \cos \theta_r = H_t \cos \theta_t$, which can be rewritten as $n_1 \cos \theta_i (1 - r) = n_2 \cos \theta_t t$ using the relation $H = nE/c$. Solving for r gives

$$r_s = \frac{n_1 \cos \theta_i - n_2 \cos \theta_t}{n_1 \cos \theta_i + n_2 \cos \theta_t} \quad (2.5)$$

while solving for t gives

$$t_s = \frac{2n_1 \cos \theta_i}{n_1 \cos \theta_i + n_2 \cos \theta_t}. \quad (2.6)$$

For p-polarized light, the magnetic fields are given by $H_i + H_r = H_t$ or $n_1(1 + r) = n_2 t$, thus the electric fields are given by $E_i \cos \theta_i - E_r \cos \theta_r = E_t \cos \theta_t$, which can be rewritten as $n_1(1 - r) \cos \theta_i = n_2 t \cos \theta_t$. Solving for r gives

$$r_p = \frac{n_2 \cos \theta_i - n_1 \cos \theta_t}{n_2 \cos \theta_i + n_1 \cos \theta_t} \quad (2.7)$$

while solving for t gives

$$t_p = \frac{2n_1 \cos \theta_i}{n_2 \cos \theta_i + n_1 \cos \theta_t}. \quad (2.8)$$

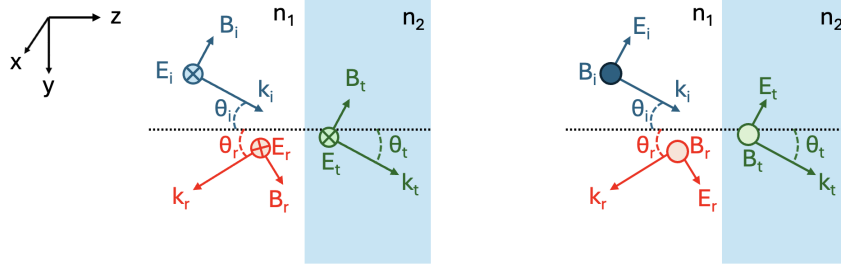


Figure 2.4: Diagram of polarized light at two interfaces. The left image shows the case for s-polarized light, with the electric field pointing into the page. For p-polarized light on the right the magnetic field points outside of the page.

At normal incidence the reflectance R evaluates to

$$R = r_1^2 = \left| \frac{n_1 - n_2}{n_1 + n_2} \right|^2. \quad (2.9)$$

For an uncoated glass substrate ($n_{\text{glass}} \approx 1.5$) in air ($n_{\text{air}} \approx 1.0$), approximately 4% of incident light power is reflected per interface (Keshavarz Hedayati Elbahri, 2016; Raut et al., 2011). For high-index semiconductor materials like silicon or GaAs ($n_{\text{Si}} \approx 3.7$ at 780 nm), reflection losses exceed 30% (Keshavarz Hedayati Elbahri, 2016).

Consider a single dielectric film of thickness d_2 and refractive index n_2 deposited between an ambient medium n_1 (air) and a substrate n_3 , where $n_1 < n_2 < n_3$ as shown in Figure 2.5. The reflectance of the structure at normal incidence would be given by

$$R = \frac{r_{12}^2 + r_{23}^2 + 2r_{12}r_{23} \cos(2\delta_2)}{1 + r_{12}^2 + r_{23}^2 + 2r_{12}r_{23} \cos(2\delta_2)} \quad (2.10)$$

where $\delta_2 = k_2 d_2 = 2\pi n_2 d_2 / \lambda$ is the phase difference between the reflected light at the interfaces of the dielectric film. The $2r_{12}r_{23} \cos(2\delta_2)$ term oscillates, and at $\cos(2\delta_2) = \pm 1$ or when $2\delta_2$ is an integer multiple of 2π , interference of the reflected waves occur and the reflectance is at an extremum. The wavelength at the extremum is

$$\lambda = \frac{2n_2 d_2}{m} \quad (2.11)$$

where m is the number of cycles from one extremum to another. The refractive index

of the thin film can be calculated if the thickness is known

$$n_2 = \frac{m\lambda_1\lambda_m}{2d_2(\lambda_m - \lambda_1)}. \quad (2.12)$$

and the new λ is the average of the two wavelengths $\lambda = (\lambda_m + \lambda_1)/2$.

The mechanism of classical thin-film ARCs relies on destructive wave interference (Raut et al., 2011). Two necessary conditions must be satisfied simultaneously to achieve zero reflectance ($R = 0$) at a specific target wavelength λ_t . The first condition is that the optical path difference between the partial wave reflected from the top interface (r_1) and the partial wave reflected from the substrate interface (r_2) must result in a π phase shift (180° out of phase). The physical film thickness d must satisfy:

$$d = \frac{\lambda_t}{4n_2}. \quad (2.13)$$

Since both reflection events occur at interfaces transitioning from a lower to a higher refractive index ($n_1 < n_2$ and $n_2 < n_3$), both undergo an phase change upon reflection. The round-trip optical path within the coating $2n_2d_2 = \lambda_t/2$ introduces a net phase change of π , ensuring destructive interference.

The second condition for zero reflectance is that the intensities of the two reflected waves must be identical $|r_1| = |r_2|$. Equalizing the Fresnel reflection coefficients yields:

$$\frac{n_2 - n_1}{n_2 + n_1} = \frac{n_3 - n_2}{n_3 + n_2} \implies n_2 = \sqrt{n_1 n_3}. \quad (2.14)$$

For a glass substrate ($n_3 = 1.5$) in air ($n_1 = 1.0$), the ideal coating index is $n_2 \approx 1.22$ (Keshavarz Hedayati Elbahri, 2016). However, finding natural bulk dielectric materials with $n \approx 1.22$ is difficult; magnesium fluoride (MgF_2 , $n \approx 1.38$) is usually employed as an alternative (Keshavarz Hedayati Elbahri, 2016). Single-layer ARCs suffer from multiple limitations. First, destructive interference is exact only at the design wavelength λ_0 . At adjacent wavelengths the reflection increases rapidly. Another limitation of single-layer ARCs is that off-normal angles of incidence alter both the

internal propagation path length and the polarization-dependent reflection coefficients, shifting the minimum reflection dip away from λ_t .

To achieve broadband and wide-angle antireflection, multi-layer dielectric stacks comprising alternating high-index and low-index thin films are used (Astrauskytė et al., 2023). Multi-layer structures introduce multiple internal reflection pathways that result in broader and more easily tailored destructive interference minima across the relevant spectrum. The design of multi-layer ARCs is usually performed computationally with the transfer matrix method[32].

2.3 Transfer Matrix Method

At the boundary between two dielectric materials, the electric field amplitudes of the incident, reflected, and transmitted waves can be related using a transfer matrix. In the schematic shown in Figure 2.5a, a wave with amplitude E_1^+ is moving to the right, and another wave with amplitude E_1^- is moving to the left. For the other side of the boundary, the amplitudes are E_2^+ and E_2^- . The fields can be characterized using two 2×1 matrices, one for each side of the boundary, and a 2×2 interface matrix $\mathbf{D}$ as shown:

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \begin{bmatrix} D_{12} & D_{21} \\ D_{12} & D_{21} \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix} \quad (2.15)$$

The values of the matrix elements D_{ij} can be obtained from inspecting Figure 2.5a. The subscripts i and j refer to the direction of the wave propagation, with t_{12} signifying transmission from medium 1 to medium 2, and r_{12} signifying reflection from medium 1 to medium 2. The fields are thus given by

$$E_2^+ = t_{12}E_1^+ + r_{21}E_2^- \quad (2.16)$$

$$E_1^- = r_{12}E_1^+ + t_{21}E_2^- \quad (2.17)$$

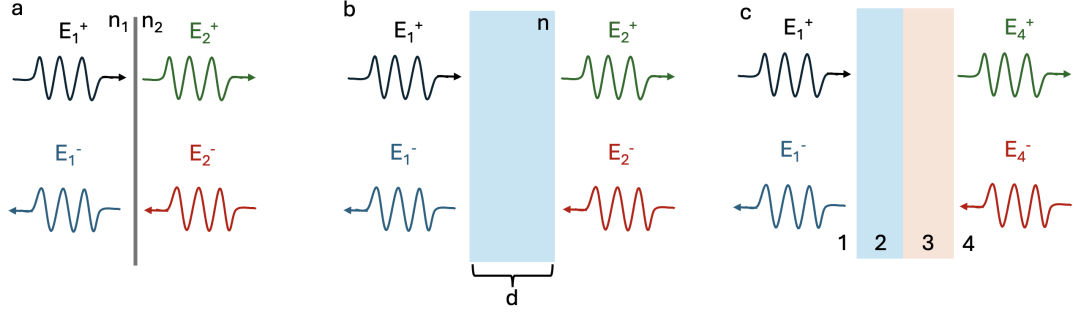


Figure 2.5: Propagation of waves at an interface (a), through a layer with refractive index n and thickness d (b), and through a multilayer system (c).

which can be rearranged to

$$E_1^+ = \frac{E_2^+ - r_{21}E_2^-}{t_{12}} \quad (2.18)$$

$$E_1^- = \frac{r_{12}}{t_{12}} (E_2^+ - r_{21}E_2^-) + t_{21}E_2^-. \quad (2.19)$$

Considering that the reflection coefficient is $r = (n_1 - n_2) / (n_1 + n_2)$ then $r_{12} = -r_{21}$. For the transmission coefficient $t = 2n_1 / (n_1 + n_2)$, and $t_{12} = n_2 t_{21} / n_1$. Substituting these relations into Equation 2.18 and Equation 2.19 with the assumption that $R + T = 1$ yields the matrix equation for the interface **D**

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \frac{1}{t_{12}} \begin{bmatrix} 1 & r_{12} \\ r_{12} & 1 \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix}. \quad (2.20)$$

The propagation matrix **P** for a layer of thickness d and refractive index n can be obtained by considering the geometry of the waves in Figure 2.5b, where a layer with refractive index n and thickness d has been inserted between the two waves. The wave moving to the right varies as $E^+ = E_0 e^{-j(k_y y + k_z z - \omega t)}$, and the difference between the two waves at the left and right boundaries is given by $E_2^+ = E_1^+ e^{j k_z \cos \theta}$ with the reverse given by $E_2^- = E_1^- e^{-j k_z \cos \theta}$. The propagation matrix is thus given by

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \begin{bmatrix} e^{j k d \cos \theta} & 0 \\ 0 & e^{-j k d \cos \theta} \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix} \quad (2.21)$$

where $k = 2\pi n / \lambda$ is the wavenumber in the medium. This analysis can be extended

to a two layer system by multiplying the interface and propagation matrices while considering the order of the layers. Starting from the rightmost boundary as shown in Figure 2.5c, the fields on both sides are related using the interface matrix by $\mathbf{E}_{3R} = \mathbf{D}_{34}\mathbf{E}_4$, where $3R$ is the field at the right side of layer 3. The fields at the left side of layer 3 are related to the right side by the propagation matrix $\mathbf{E}_{3L} = \mathbf{P}_3\mathbf{E}_{3R}$, which in turn can be related using another interface matrix to the field across the relevant boundary. This results in the following matrix equation relating the fields at the leftmost region to the rightmost region:

$$\mathbf{M} = \mathbf{D}_{12}\mathbf{P}_2\mathbf{D}_{23}\mathbf{P}_3\mathbf{D}_{34} \quad (2.22)$$

such that

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \mathbf{M} \begin{bmatrix} E_4^+ \\ E_4^- \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} E_4^+ \\ E_4^- \end{bmatrix}. \quad (2.23)$$

This method allows the analysis of any multilayer system using matrix multiplication of the component interface $\mathbf{D}$ and propagation $\mathbf{P}$ matrices. The reflection coefficient r for a multilayer structure is simply

$$r = \frac{M_{21}}{M_{11}} \quad (2.24)$$

since $t = 1/M_{11}$ and $r = M_{21}/M_{11}$.

2.4 Electron beam physical vapor deposition of a-Si and SiO₂

Silicon and its native oxide SiO₂ are the most common materials used in the semiconductor industry. Thin films of Si and SiO₂ are used in microelectronics, photonics, and photovoltaic applications. These thin films are grown using methods ranging from physical vapor deposition, chemical vapor deposition, to thermal oxidation. Electron beam physical vapor deposition (EBPVD) or e-beam evaporation is one commonly utilized technique for depositing high-purity Si and SiO₂ films due to its high deposition

rate, precise thickness control, and capability to vaporize high-melting-point materials.

SiO₂ is widely utilized due to its wide bandgap (≈ 8.9 eV), low refractive index ($n \approx 1.45 - 1.46$ in the visible spectrum), high chemical stability, and low dielectric loss for optical coatings (Macleod, 2010). While thermal oxidation can produce high-quality native SiO₂ from Si, it requires high temperatures ($>300^\circ\text{C}$) and is restricted to silicon substrates. Electron beam evaporation enables low-temperature deposition ($<300^\circ\text{C}$) on different substrates including polymers, glass, and compound semiconductors such as GaAs (Schiller et al., 1982). However, e-beam evaporating silica presents unique physical challenges—specifically dissociation during evaporation, sub-stoichiometry SiO₂, porosity, and intrinsic film stress (Ritter, 1972).

In e-beam evaporation, a high-energy electron beam ($6 - 10$ kV) is magnetically steered onto a high-purity target contained in a water-cooled copper crucible. The electron generation mechanism relies on Richardson-Dushman thermionic emission from a heated filament (typically tungsten):

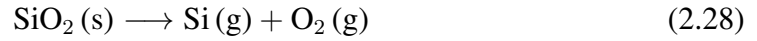
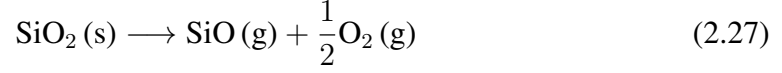
$$J = AT^2 \exp\left(-\frac{\Phi}{k_B T}\right) \quad (2.25)$$

where J is the emission current density A/m², $A \approx 1.2 \times 10^6$ A/(m² K²) is the Richardson constant, Φ is the work function of the filament material eV, k_B is the Boltzmann constant and T is the absolute temperature in K. Emitted electrons are accelerated toward an anode using a high potential difference. The beam trajectory is controlled via the Lorentz force:

$$\mathbf{F} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (2.26)$$

In a uniform transverse magnetic field $\mathbf{B}$, electrons follow a circular path. This bending caused by the magnetic field allows the electron gun to remain shielded from direct line-of-sight deposition from the source while enabling frequent sweeping across the crucible to maintain uniform evaporation from the source as shown in Figure 2.4. As the electron beam strikes the source, local temperatures at the beam spot may exceed

1700°C, causing melting and evaporation for Si or direct sublimation in the case of SiO₂ (Schiller et al., 1982). Unlike pure Si, SiO₂ does not evaporate purely as intact SiO₂ molecules. Thermal dissociation may occur at the surface of the source, yielding a mixture of gaseous species (Ritter, 1972):



Unassisted e-beam evaporation of SiO₂ in high vacuum typically yields sub-stoichiometric films of SiO_x where $x < 2.0$. Sub-stoichiometry impacts film performance by introducing sub-oxide states (Si¹⁺, Si²⁺, Si³⁺) and oxygen vacancies, leading to an increased optical absorption and increased refractive index. To ensure fully stoichiometric SiO₂, reactive e-beam evaporation can be performed by introducing oxygen gas into the chamber.

According to the structure zone model, the growth of the films proceed in three general steps (Thornton). The first step occurs when the vaporized molecules are transported towards the substrate, the second when they are adsorbed onto the surface of the substrate or growing film, their diffusion across the surface, and any chemical reactions that may involve their incorporation into the film or removal via thermal desorption or sputtering. The third step involves the movement of the molecules to their final locations via bulk diffusion and solid state reactions.

Movchan and Demchishin classified film growth into distinct zones, with the morphology of the films depending strongly on the normalized homologous temperature $T_h = T_{\text{sub}}/T_m$ where T_{sub} is the substrate temperature and T_m is the melting point of the material in K. At low temperatures ($T_h < 0.3$) surface diffusion is low, and molecules essentially stick where they arrive, producing porous, tapered columnar structures. This is considered as Zone 1. At Zone 2 ($T_h < 0.5$) the temperatures are high enough to allow the surface diffusion fluxes to dominate over the arrival fluxes, but still low enough such that the bulk diffusion rates remain orders of magnitude less than the surface dif-

fusion rates. When the temperature is high enough, bulk diffusion will dominate over all other processes and fully dense grains form. This is considered as Zone 3. The morphology of the films also has an effect on the refractive index, with porous films generally having a lower index compared to the bulk.

2.5 Envelope method for refractive index

CHAPTER 3

METHODOLOGY

3.1 Initial growth of a-Si and SiO₂ thin films via electron beam evaporation

Glass slides and Si wafers were used as substrates for the initial growth via electron beam evaporation of a-Si and SiO₂ thin films respectively. The substrates were cleaned by sonication in acetone, isopropanol, and deionized water for 5 minutes each, after which they were blow-dried with nitrogen gas. The substrates were then loaded into the ULVAC electron beam evaporation chamber, where a base pressure of 4.5×10^{-4} Pa was achieved before actual deposition. Deposition was carried out at room temperature, and the nominal deposition rates and thicknesses of the films were monitored using a quartz crystal microbalance (QCM) with the input values in Table 3.1.

3.2 Characterization of a-Si and SiO₂ thin films

After unloading the samples from the chamber, the thicknesses of the films were measured using x-ray reflectivity (XRR) with a Malvern Panalytical Empyrean x-ray diffractometer. Curve fitting was performed on selected samples to obtain the densities and interface roughness of the films. The refractive index of the a-Si film was determined using Swanepoel's method via UV-Vis spectroscopy with a Shimadzu UV-Vis spectrophotometer while the refractive index of the SiO₂ film was identified using the interference method via the reflectance spectroscopy setup as shown in Figure 3.1. The measured refractive indices of the a-Si and SiO₂ films were used for the design of the a-Si/SiO₂ ARC stack.

	z-factor	ρ g/cm ³
Si	0.712	2.32
SiO ₂	1	2.65

Table 3.1: QCM inputs

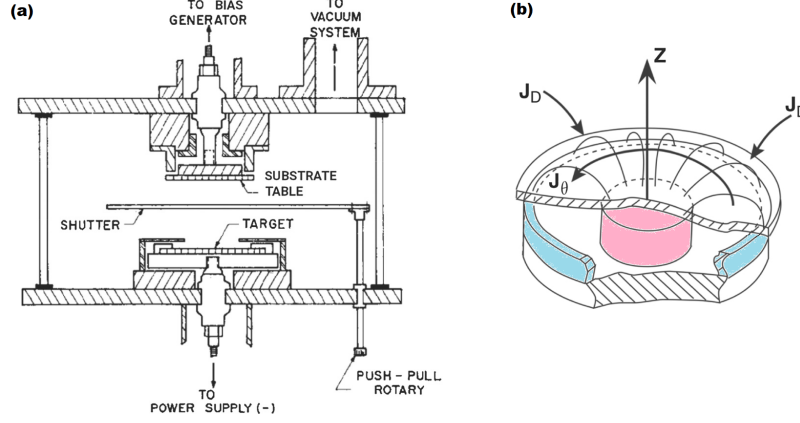


Figure 3.1: Schematics of a magnetron sputtering system (a) with cathode target (b). The plasma is confined by the magnetic field in a toroidal pattern[4].

3.3 Simulation and growth of a-Si/SiO₂ antireflective coatings

A transfer matrix method (TMM) implementation in Python was used to calculate the optimal thicknesses of the Si and SiO₂ layers for the antireflective coating stack. The refractive indices of the a-Si and SiO₂ layers used for the simulation were obtained from the previous section. The optimal thicknesses of the layers were calculated to minimize the reflectance at 1550 nm and 780 nm, which are the wavelengths of the lasers which can be used in the THz-TDS spectroscopy system. The program iterated over a range of thicknesses for each layer while calculating the reflectance of the stack, selecting the thicknesses that resulted in the lowest reflectance at the target wavelengths. The thicknesses obtained from the simulation were then used for the growth of the a-Si/SiO₂ ARC on bare wafers via electron beam evaporation. The reflectance of the samples with the antireflective coating was measured via reflectance spectroscopy and compared to the simulated reflectance to verify the accuracy of the simulation and growth process.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Thicknesses and densities of a-Si and SiO₂ thin films

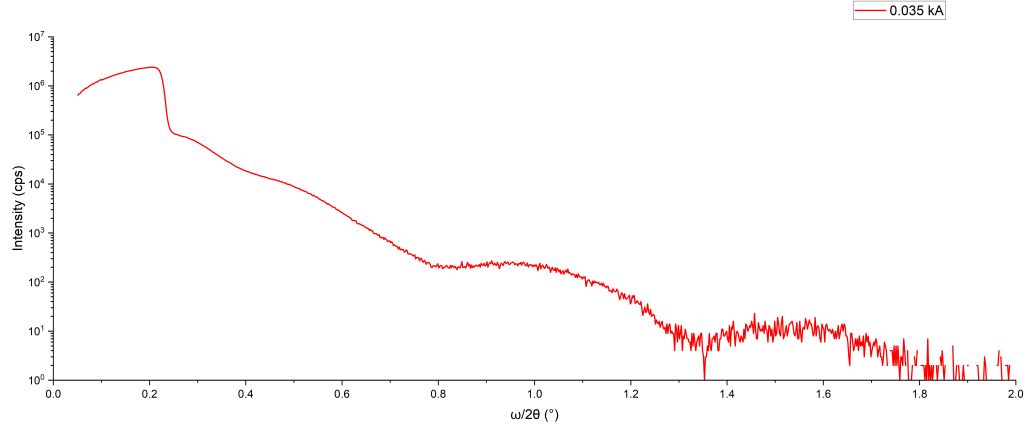
Figure 4.1 and Figure 4.2 show the XRR measurements of various a-Si films grown on glass substrates and SiO₂ films grown on Si(100) substrates respectively with varying thicknesses according to the QCM readings across different growth runs. From the graphs of the intensity the 2θ range at which the detector can count the reflected x-rays generally ranges from 0.3° to 2.0° . The measured thicknesses of the a-Si film based on XRR are summarized in Table 4.1, while for SiO₂ the results are summarized in Table 4.2. No thickness data could be obtained from XRR for films which were thicker than $1.52 \text{ k}\text{\AA}$ for a-Si and $1.369 \text{ k}\text{\AA}$ for SiO₂.

Figure 4.5, Figure 4.6, Figure 4.7 and Figure 4.8 show the simulated curves using the XRR software for selected a-Si films on glass substrates and SiO₂ films on Si. The values of thickness, interface roughness and densities obtained by fitting the curve are summarized in Table 4.3.

Linear regression of the measured thicknesses versus the QCM readings for a-Si as shown in Figure 4.3 yields the equation

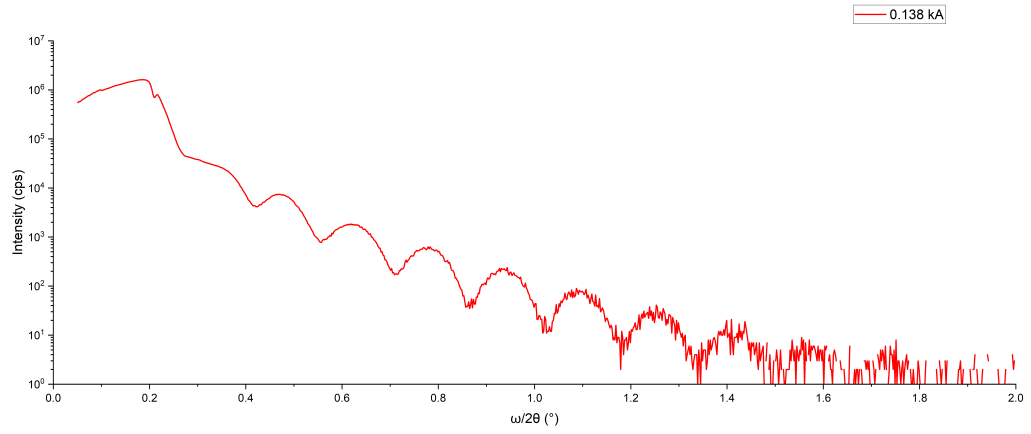
QCM (k $\text{\AA}$)	Direct (nm)	Fourier (nm)
0.035	7.822	6.90
0.069	14.077	13.80
0.104	21.290	21.20
0.138	27.529	27.50
1.270	214.620	213.15
1.520	259.840	259.20

Table 4.1: a-Si thicknesses measured using the QCM, direct fringe calculation and Fourier method from the XRR spectra



(a) 0.035 kÅ a-Si thin film

Figure 4.1: XRR measurements of a-Si thin films with varying QCM thicknesses



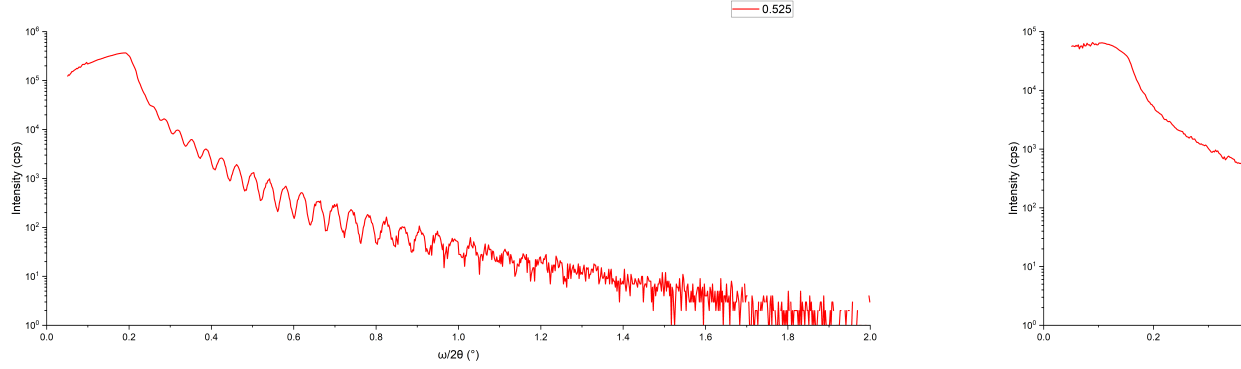
(d) 0.138 kÅ a-Si thin film

Figure 4.1: XRR measurements of a-Si thin films with varying QCM thicknesses, continued

$$y = 168.03x + 3.0403 \quad (4.1)$$

with $R^2 = 0.9999$, where y is the measured XRR thickness in nm and x is the QCM reading in kÅ. Similarly, linear regression of the measured XRR thickness and the QCM values for SiO₂ as shown in Figure 4.4 yields the following equation with $R^2 = 0.9941$:

$$y = 198.52x - 4.5056 \quad (4.2)$$



(a) 0.525 kÅ SiO₂ thin film

Figure 4.2: XRR measurements of SiO₂ thin films with varying QCM thicknesses

QCM (kÅ)	Direct (nm)	Fourier (nm)
0.525	104.6	103.9
0.884	165.1	163.2
1.050	206.6	204.7
1.369	272.3	270.1

Table 4.2: SiO₂ thicknesses measured using the QCM, direct fringe calculation and Fourier method from the XRR spectra

	Thickness (nm)	Roughness (nm)	Density (g/cm ³)
21.20 nm a-Si	21.9115	0.7047	1.80726
27.50 nm a-Si	28.4339	0.8008	1.75748
103.9 nm SiO ₂	104.5311	0.3626	1.69738
204.7 nm SiO ₂	203.8994	0.741	2.04566

Table 4.3: Thickness, roughness and densities calculated from XRR curve fitting

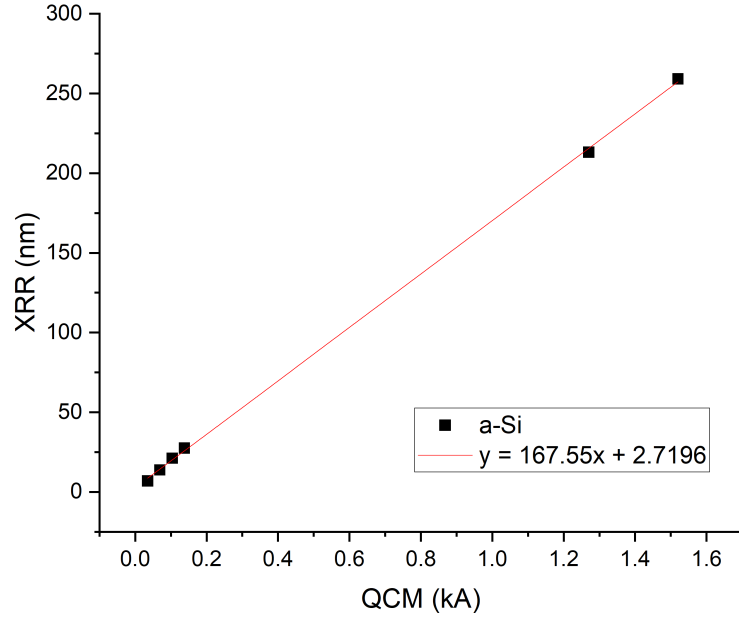


Figure 4.3: Linear regression of XRR and QCM thicknesses for a-Si thin films grown via e-beam evaporation.

The initial comparison of the QCM readings and the measured thicknesses during XRR indicate that the QCM readings during deposition generally underestimate the actual thicknesses of the films. This underestimation is likely due to the mismatch in densities used as input for the QCM as shown in Table 3.1 and the actual densities of the evaporated film as shown in Table 4.3. The thicknesses of the films as obtained by curve fitting generally agree with the direct and Fourier method of analysis for the Kiessig fringes. The low interface roughness of the films imply that electron beam evaporation is an effective method to produce uniform films, albeit with much lower densities than the bulk material. This discrepancy may be caused by the increased porosity of the film

Nevertheless, the results of the regression analysis indicate that the QCM readings scale linearly with the actual thicknesses of the films as measured by XRR. The high R^2 values suggest that the QCM is a reliable tool for monitoring film thickness during deposition as long as the linear relationship is considered, and that the linear relationships obtained in this study can be used to calibrate the QCM readings for future depositions. The lower R^2 value in the regression for SiO_2 is attributed to the variation in the de-

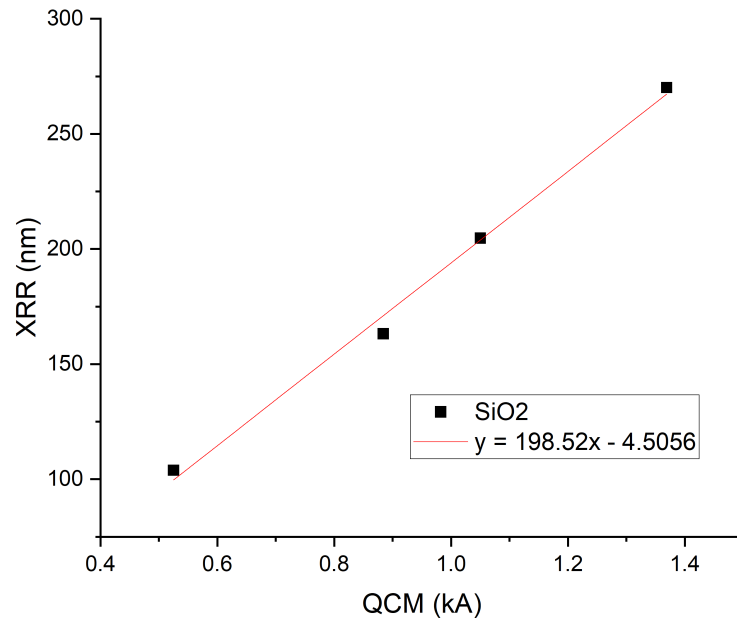


Figure 4.4: Linear regression of XRR and QCM thicknesses for SiO₂ thin films grown via e-beam evaporation.

position rate during the growth of the SiO₂ films, which is caused by the difficulty in maintaining a stable deposition rate for SiO₂ due to its low thermal conductivity and tendency to sublime instead of melting into a uniform liquid pool during heating by the electron beam as is the case for other materials.

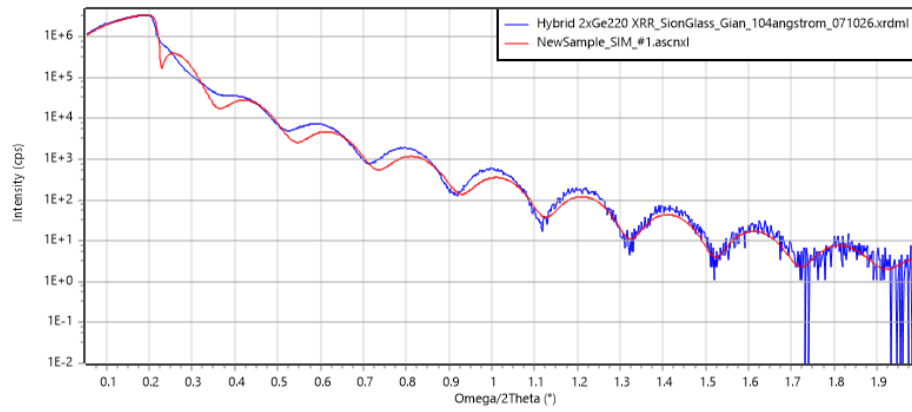


Figure 4.5: XRR curve fit to 21.20 nm a-Si thin film

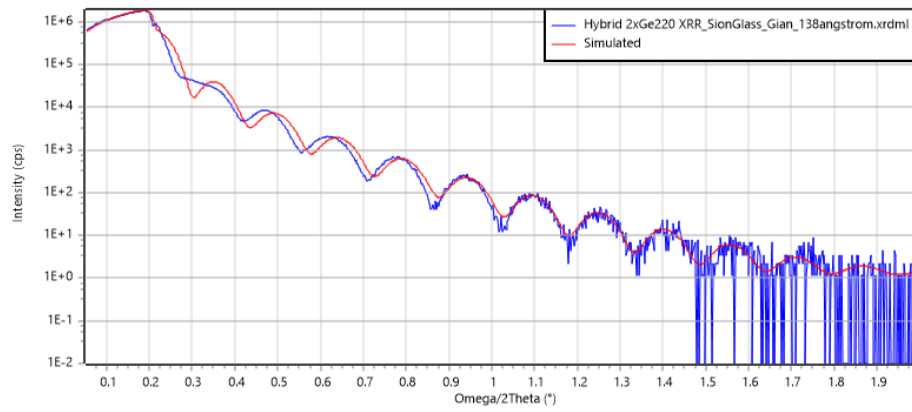


Figure 4.6: XRR curve fit to 27.50 nm a-Si thin film

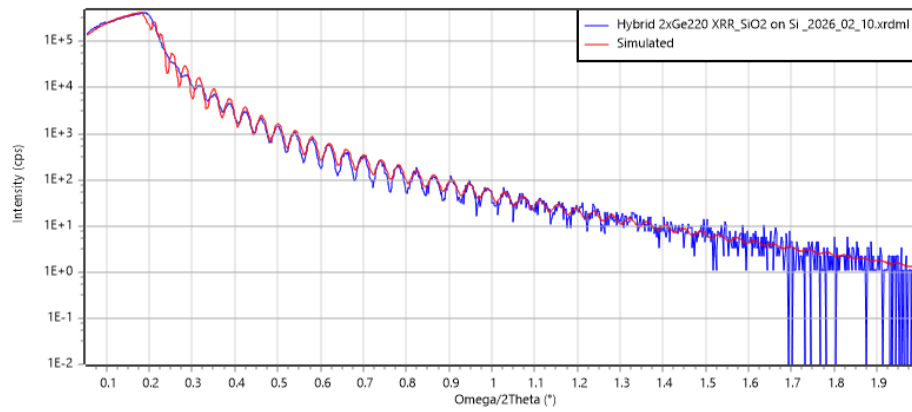


Figure 4.7: XRR curve fit to 103.9 nm SiO₂ thin film

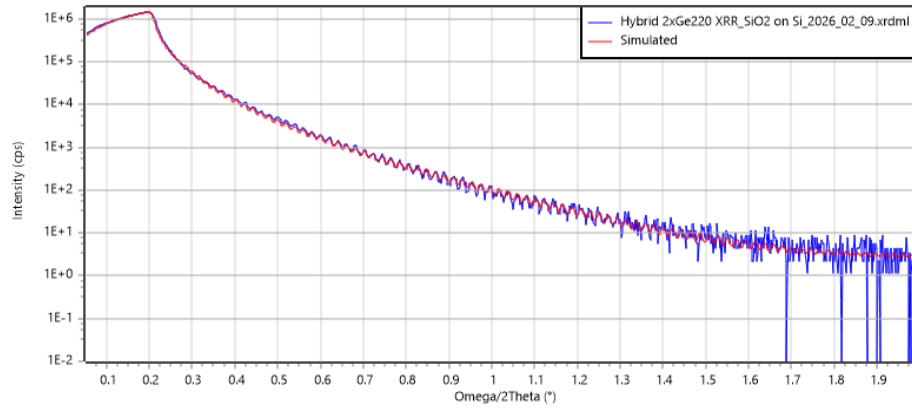


Figure 4.8: XRR curve fit to 204.7 nm SiO₂ thin film

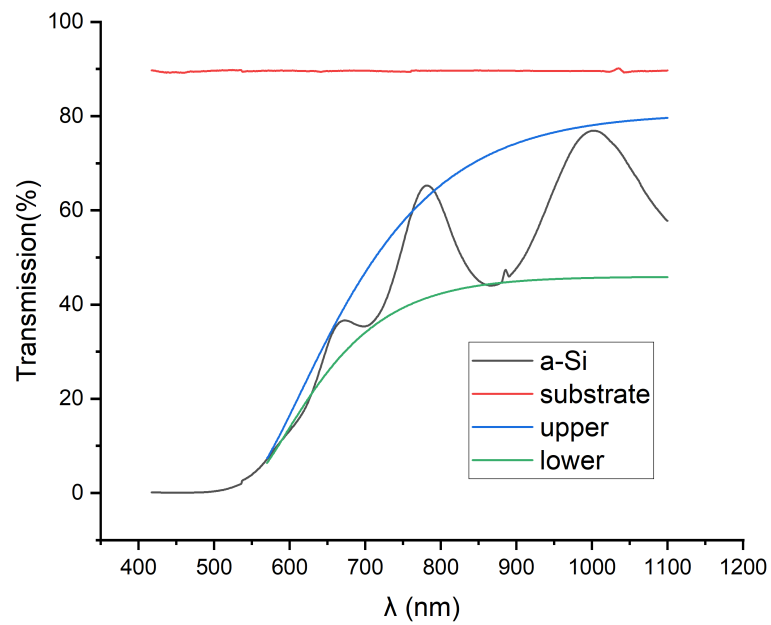


Figure 4.9: The transmission curves for the 600 nm a-Si thin film and the glass substrate together with the upper and lower envelopes for the envelope method

4.2 Refractive index of a-Si thin films

Figure 4.9 shows the UV-Vis transmittance spectra of a 600 nm a-Si thin film deposited on a glass substrate together with the upper and lower envelopes as calculated using the envelope method as described in section 2.5. The equations of the exponential functions fitted to the extrema points listed in Table 4.4 are

$$y = 80.612419 + 1.200883 \times 10^9 e^{-x/89.819343} - 12.00749 \times 10^9 e^{-x/89.821415} \quad (4.3)$$

for the upper envelope with $R^2 = 0.9969$ and

$$y = 45.9178758 + 3.6028012 \times 10^{10} e^{-x/64.3969697} - 3.16015549 \times 10^{10} e^{-x/64.3973202} \quad (4.4)$$

for the lower envelope with $R^2 = 0.9977$. The equation for the refractive index of the a-Si thin film according to Sellmeier's equation fitted to the data points of the refractive indices obtained from the envelope method is

$$n^2 = 14.84529 + \frac{7.06709 \times 10^{12} \lambda^2}{\lambda^2 + 2.12449 \times 10^{18}} + \frac{-10.41841 \lambda^2}{\lambda^2 + 190469.027}. \quad (4.5)$$

and the graph of the resulting refractive index versus wavelength is shown in Figure 4.10. The average thickness of the a-Si thin film according to the envelope method is 653 nm.

The refractive index of the a-Si thin film is comparable to the literature values obtained by Swanepoel for a-Si, with values ranging from 3.0 to 3.2 across the relevant wavelengths. However, the refractive index does not decrease monotonically with increasing wavelength as expected for the material. This deviation from the expected behavior is unlikely to be anomalous dispersion and instead is probably caused by the few number of data points from the envelope method used to fit the Sellmeier equation. This is supported by the difference in calculated thickness for the film using the envelope method and the extrapolation using Equation 4.1. A larger number of oscilla-

λ (nm)	T_{film}	T_{sub}	T_M	T_m	n_s	N	n_1	d_1 (nm)	m	d_2 (nm)	n_2
maxima											
1002	76.9	86.3	78.1	45.7	1.62	4.76	3.04	698.1	3.47	659.4	3.07
782	65.3	88.2	62.9	41.4	1.61	4.45	2.93	422.4	4.28	666.8	3.00
655	33.5	89.5	34.3	26.7	1.61	5.34	3.23	564.0	5.64	608.1	3.01
590	11.3	89.7	13.3	11.3	1.61	6.00	3.43				
minima											
871	44.1	86.9	72.3	44.5	1.61	4.59	2.98	737.9	3.91	657.6	3.00
709	36.1	89.6	49.0	35.2	1.61	4.38	2.91	426.7	4.68	670.8	2.99
613	16.4	89.6	21.5	17.7	1.61	5.34	3.23	281.5*			
571	7.4	89.6	7.6	6.6	1.61	8.18	4.02				
$\bar{d}$								569.8		652.53	

Table 4.4: Relevant values for the envelope method of the a-Si thin film.* The last d_1 of 281.5 nm is excluded from further calculations due to its distance from the other thickness values

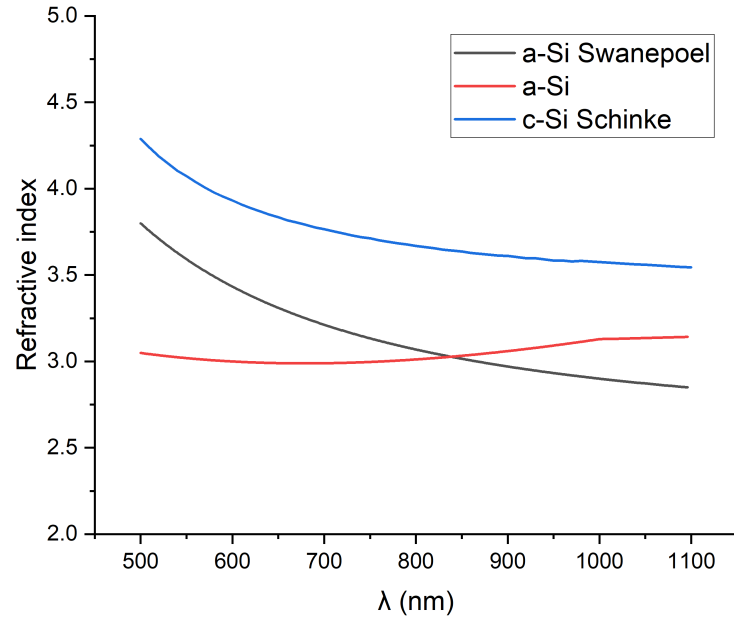


Figure 4.10: Refractive index of a-Si thin film from 500 nm to 1100 nm

tions and data points would likely yield a more accurate Sellmeier equation that would exhibit the proper dispersion behavior across the wavelength range[40]. Furthermore, the a-Si thin film refractive index is much lower than that of crystalline silicon, which has a refractive index of 3.5 to 4.25 across the range of wavelengths as measured by Schinke[41]. This is likely due to the lower density and less ordered structure of the a-Si thin film compared to crystalline silicon, and supports the previous finding in section 4.1.

4.3 Refractive index of SiO₂ thin films

Figure 4.11 shows the reflectance spectra of a 990 nm SiO₂ thin film deposited on a Si substrate after deconvolution using a mirror. The extrema points used for the calculation of the refractive index of the film using Equation 2.12 are listed in Table 4.5 together with the corresponding refractive indices and the new wavelengths. Figure 4.12 shows the graph of the refractive index after fitting the data points to Sellmeier's equation, which is

$$n^2 = 14.84529 + \frac{7.06709 \times 10^{12} \lambda^2}{\lambda^2 + 2.12449 \times 10^{18}} + \frac{-10.41841 \lambda^2}{\lambda^2 + 190469.027}. \quad (4.6)$$

with $R^2 = 0.998$.

The refractive index of the SiO₂ thin film is comparable to the literature values obtained from Gao[42]. The slight increase in refractive index versus the literature

λ_0 (nm)	R	λ_1 (nm)	n_1	λ_2 (nm)	n_2
minima					
530.0	0.052	587.5	1.50	676.3	1.50
645.0	0.040	733.8	1.50		
822.5	0.043				
maxima					
577.5	0.154	648.8	1.47	765.0	1.48
720.0	0.140	836.3	1.48		
952.5	0.216				

Table 4.5: Refractive indices and wavelengths for the SiO₂ thin film

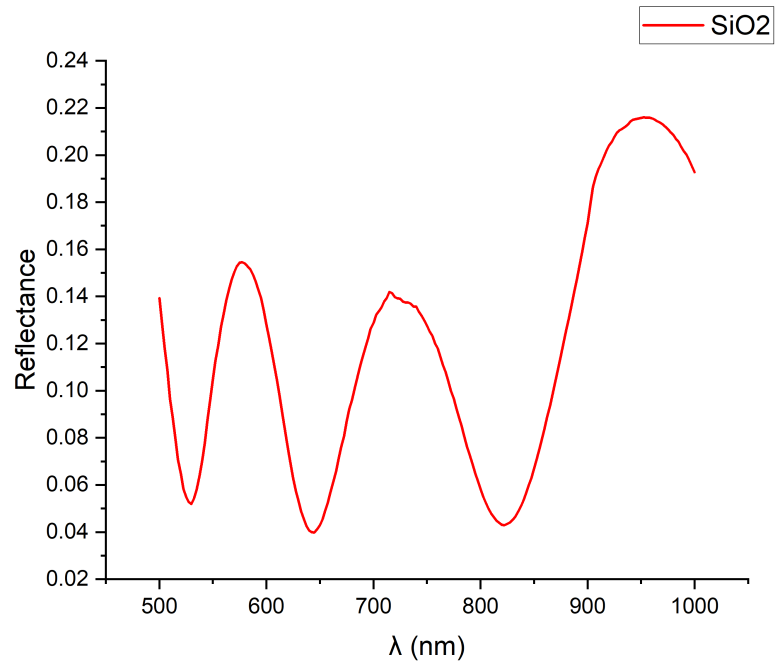


Figure 4.11: Deconvoluted reflectance spectra of 990 nm SiO₂ thin film

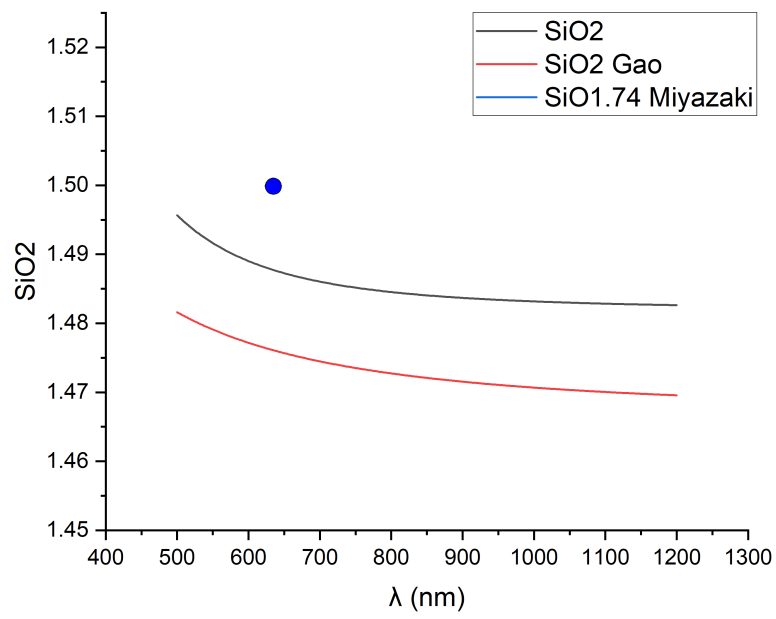


Figure 4.12: Refractive index of the SiO₂ film compared to Gao and Miyazaki

values could be attributed to the stoichiometry of the SiO_2 thin film caused by a lack of reactive gas during the deposition process, leading to a higher refractive index closer to that of silicon. Miyazaki et al. deposited $\text{SiO}_{1.74}$ thin films and found that the refractive index at 633 nm was 1.50, which is higher than the refractive index. The lower density of the SiO_2 thin film compared to bulk SiO_2 as measured by XRR previously may be caused by the increased porosity of the film or the decreased ratio of oxygen versus silicon.

4.4 Reflectance of a-Si/ SiO_2 antireflective coatings

Figure 4.13 shows the reflectance spectra of a-Si/ SiO_2 antireflective coatings with layer thicknesses of 53 nm/14 nm/206 nm, 48 nm/19 nm/208 nm, and 49 nm/24 nm/251 nm for the SiO_2 /a-Si/ SiO_2 layers respectively using the refractive indices obtained in the previous sections. The simulated reflectance spectra for the first ARC shows a value less than 1 % around 630 nm, whereas the simulation of the second ARC shows a minimum at 660 nm. The measured reflectance spectra for the ARCs show that the minimum value around 4 % for the first ARC at 660 nm and a minimum value of 2.5 % for the second ARC at 720 nm. Table 4.6 summarizes the relevant information of the ARCs being studied. The blank Si wafer exhibits a reflectivity ranging from 50 % to 75 % across the wavelength range of 450 nm to 1000 nm.

The layers deposited using electron beam deposition demonstrated a reduced value of reflectance compared to the bare wafer, proving their effectiveness as antireflective

	1st zero		2nd zero	
	λ (nm)	R	λ_0 (nm)	R
ARC 1 TMM				
ARC 1				
ARC 2 TMM				
ARC 2				
ARC 3 TMM				
ARC 3				

Table 4.6: Thickness, roughness and densities calculated from XRR curve fitting

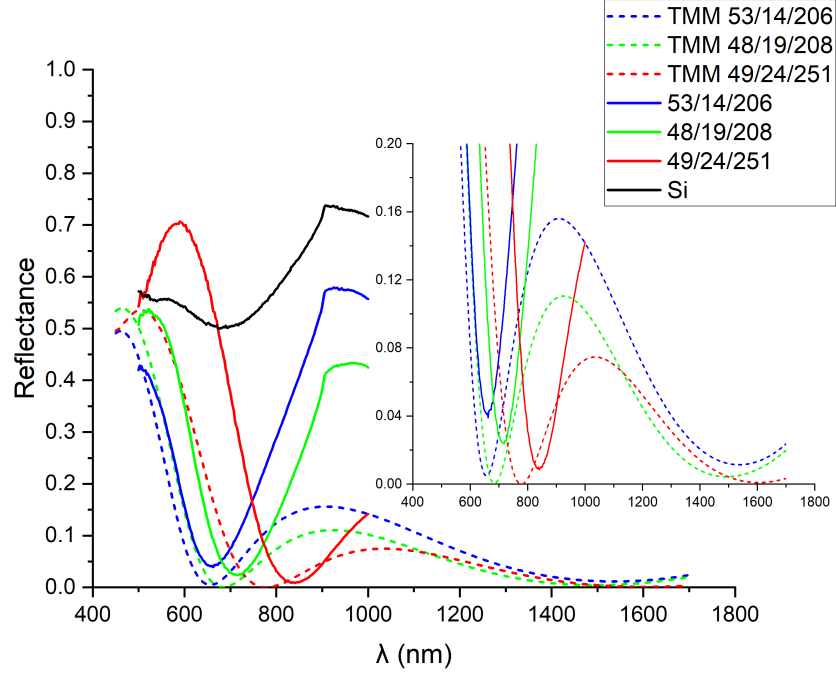


Figure 4.13: Reflectivities of ARCs with varying layer thicknesses simulated using the transfer matrix method and measured experimentally. The reflectivity of a bare Si wafer is included for comparison.

coatings. The mismatch in the minimum values obtained experimentally versus the simulation is due to the limitation of the experimental setup, which measures the reflectance of the sample relative to the mirror. More accurate reflectivity values would be obtained from absolute measurements which consider the total amount of light incident and reflected from the sample. The deviation of wavelengths at which the measured reflectance exhibit the minima from the simulated values can be attributed to the inaccuracy in measuring the refractive index of the thin films which were previously deposited, especially for a-Si.

CHAPTER 5

CONCLUSION

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Gantt chart

[illegible]

ITEM	DESCRIPTION	UNIT COST	QUANTITY	TOTAL
Fabrication supplies				
Acetone	4L (JTBaker)	2,300.00	1 bottle	2,300.00
38% Hydrochloric acid	2.5L (JTBaker)	3,955.00	1 bottle	3,955.00
Photoresist	250ml (ma-P 1225)	37,200.00	1 bottle	37,200.00
Developer	2.5L (mr-D 526/S)	24,100.00	1 bottle	24,100.00
Si wafers	Si(100), 2", undoped	11,342.50	2 pcs	22,685.00
Others				
Printing				5,000.00
Characterization	Scanning electron microscopy			10,000.00
Total				105,240.00